1. Using data starting with 1950 and projected to 2050, the number of women (in millions) in the workforce is given by the function $P = 0.79x + 20.86$, where x equals the number of years after 1950.

(a) Find the value of P when $x = 36$. Explain what this means.
(b) What value of x represents 2020? Use the model to find the number of women in the workforce in 2020.

(**a**) The value of P when $x = 36$ is ______ million.
(Type an integer or decimal rounded to two decimal places as needed.)

What is the meaning of this value of P?

- **A.** The value of P represents the number of women in the workforce in the year 1936.
- **B.** The value of P represents the number of women who will join the work force between the years 1900 and 1936.
- **C.** The value of P represents the number of women who will join the work force between the years 1950 and 1986.
- **D.** The value of P represents the number of women in the workforce in the year 1986.

(b) The value of x that represents 2020 is x = ______.

The number of women in the workforce in 2020 will be ______ million.
(Type an integer or decimal rounded to two decimal places as needed.)

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2. The amount of federal tax per capita (per person) can be modeled by the function $S = 5.2707t^3 - 55.19t^2 + 579.61t + 5179.2$ where t is the number of years after 1996.

a. What are the values of t that correspond to 2005, 2011, and 2020?
b. $S = f(3)$ gives the value of S for what year? What is f(3)?
c. What x_{min} and x_{max} should be used to set a viewing window so that t represents 1996–2020?

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a. What value of t corresponds to the year 2005?

t = ______

What value of t corresponds to the year 2011?

t = ______

What value of t corresponds to the year 2020?

t = ______

b. $S = f(3)$ gives the value of S for what year?

______

What is f(3)?

f(3) = ______
(Round to the nearest integer as needed.)

c. What x_{min} should be used for the viewing window?

x_{min} = ______

What x_{max} should be used for the viewing window?

x_{max} = ______

*3. Using a graphing calculator, estimate the real zeros, the relative maxima and minima, and the range of the polynomial function.

$g(x) = x^3 - 2.7x + 3$

The zero(s) of the function is/are at approximately x = ☐.
(Round to three decimal places as needed. Use a comma to separate answers as needed.)

Find the relative maxima. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The function has a relative maximum of ☐ at $x \approx$ ☐.
  (Round to three decimal places as needed.)
- **B.** The function does not have any relative maxima.

Find the relative minima. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The function has a relative minimum of ☐ at $x \approx$ ☐.
  (Round to three decimal places as needed.)
- **B.** The function does not have any relative minima.

The range of the function is ☐.
(Type your answer in interval notation.)